

EDC BLOCK-003 (Derivation v9):

NC-2 Attempt: DGP / Induced Gravity

Can Brane-Localized Curvature Break the Degeneracy?

EDC Working Document

2026-02-02 — Pivot attempt after NC-1 inconclusive

Abstract

Derivation v8 showed that NC-1 (zero-mode normalization) yields an identity rather than an independent equation: C and σ always appear in the same combination. This note tests NC-2: Dvali-Gabadadze-Porrati (DGP) induced gravity, where a brane-localized Einstein-Hilbert term provides an independent 4D Planck mass M_4 distinct from the bulk-derived contribution. **Outcome:** INCONCLUSIVE. DGP introduces a new scale $r_c = M_5^3/M_4^2$ (crossover radius), but M_4^2 itself depends on unknown parameters (N , Λ_{UV} , regularization). The degeneracy is not broken; it is relocated from C to the induced term coefficient. BLOCK-003 remains [OPEN].

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1 Motivation: Why NC-2 After NC-1?

From v8, NC-1 (canonical KK reduction) produced:

$$G_N = \frac{C\sigma^{-3/4}}{8\pi R_\xi} \quad (1)$$

where C and σ are degenerate — they always appear together. This is a *reparametrization*, not an independent equation.

NC-2 hypothesis: In DGP-type models, 4D gravity emerges from a brane-localized curvature term, providing an *independent* scale M_4 that could break the C - σ degeneracy.

2 DGP Setup

2.1 Action

The DGP action combines bulk and brane Einstein-Hilbert terms:

$$S = \underbrace{\frac{M_5^3}{2} \int_{\mathcal{M}^5} d^5 X \sqrt{-g^{(5)}} R^{(5)}}_{S_{\text{bulk}}} + \underbrace{\frac{M_4^2}{2} \int_{\Sigma^4} d^4 x \sqrt{-h} R^{(4)}}_{S_{\text{brane}}} + S_{\text{matter}} \quad (2)$$

where:

- $M_5^3 = 1/\kappa_5^2$ is the 5D Planck mass cubed
- M_4^2 is the 4D brane-localized Planck mass squared
- $R^{(5)}$, $R^{(4)}$ are 5D and 4D Ricci scalars

2.2 Key Feature: Two Independent Scales

Unlike pure KK reduction (NC-1), DGP has *two* independent mass scales:

- M_5 : bulk gravitational scale
- M_4 : brane gravitational scale

The crossover radius is:

$$r_c = \frac{M_5^3}{M_4^2} \quad (3)$$

At distances $r \ll r_c$: 4D gravity dominates ($G_N \sim 1/M_4^2$).

At distances $r \gg r_c$: 5D gravity dominates (leakage into bulk).

3 Effective 4D Newton Constant

3.1 DGP Graviton Propagator

The effective 4D gravitational coupling at distance r is:

$$G_N^{\text{eff}}(r) = \frac{1}{8\pi M_4^2} \cdot F(r/r_c) \quad (4)$$

where $F(x) \rightarrow 1$ as $x \rightarrow 0$ (4D regime).

In the 4D regime ($r \ll r_c$):

$$\boxed{G_N = \frac{1}{8\pi M_4^2}} \quad (5)$$

3.2 Comparison to NC-1

In NC-1 (pure KK):

$$G_N^{\text{NC1}} = \frac{\kappa_5^2}{8\pi L} = \frac{1}{8\pi M_5^3 L} \quad (6)$$

In NC-2 (DGP):

$$G_N^{\text{NC2}} = \frac{1}{8\pi M_4^2} \quad (7)$$

Key difference: In DGP, G_N is determined by M_4 , not by M_5 and L . This *could* break the degeneracy if M_4 is independently specified.

4 Where Does M_4 Come From?

4.1 Induced Gravity Mechanism

In Sakharov/Adler induced gravity, M_4^2 arises from quantum loops of brane-localized matter:

$$M_4^2 = \frac{N}{16\pi^2} \Lambda_{\text{UV}}^2 \cdot f(\text{reg}) \quad (8)$$

where:

- N : number of matter species on the brane
- Λ_{UV} : UV cutoff
- $f(\text{reg})$: regularization-dependent $\mathcal{O}(1)$ factor

4.2 EDC Interpretation

For EDC, possible identifications:

- $\Lambda_{\text{UV}} = \sigma^{1/4}$ (tension-derived cutoff) [P]
- $\Lambda_{\text{UV}} = R_\xi^{-1}$ (weak scale cutoff) [P]
- $N = ?$ (unknown; EDC does not specify species count)

With $\Lambda_{\text{UV}} = \sigma^{1/4}$:

$$M_4^2 \sim \frac{N}{16\pi^2} \sigma^{1/2} \quad (9)$$

Then:

$$G_N = \frac{1}{8\pi M_4^2} \sim \frac{16\pi^2}{8\pi N \sigma^{1/2}} = \frac{2\pi}{N \sigma^{1/2}} \quad (10)$$

5 Does NC-2 Break the Degeneracy?

Proposition 1 (NC-2 Degeneracy Relocation). *In DGP/induced gravity, the unknown constant C from NC-1 is replaced by the unknown species count N (and regularization factor f). The degeneracy is relocated, not eliminated.*

Proof. From Eq. (10):

$$G_N \sim \frac{2\pi}{N \sigma^{1/2}}$$

To predict G_N , we need N and σ . Neither is specified by EDC:

- σ : brane tension (same unknown as NC-1)
- N : species count (new unknown, replaces C)

The product $N \sigma^{1/2}$ is degenerate, just as $C \sigma^{3/4}$ was in NC-1. □

5.1 Comparison of Degeneracies

Approach	G_N Formula	Unknown Combination
NC-1 (KK)	$C\sigma^{-3/4}/(8\pi R_\xi)$	$C\sigma^{-3/4}$
NC-2 (DGP)	$2\pi/(N\sigma^{1/2})$	$N\sigma^{1/2}$

Table 1: Both approaches have one degenerate combination.

6 What Would Fix NC-2?

For NC-2 to close BLOCK-003, EDC would need to specify:

1. **Species count N :** How many field species live on the brane? EDC's Z_6 structure suggests discrete particle content, but N is not derived.
2. **Regularization scheme:** The factor $f(\text{reg})$ is scheme-dependent. A canonical choice (e.g., dimensional regularization) would fix this to $\mathcal{O}(1)$.
3. **Brane tension σ :** Same requirement as NC-1.

Minimal fix: If N is small and known (e.g., $N = 1$ for a single Plenum field), and σ is specified, then G_N becomes predictable.

7 Outcome

Outcome

Status: INCONCLUSIVE

What was achieved:

- DGP framework applied to EDC [DC]
- $G_N = 2\pi/(N\sigma^{1/2})$ derived (with $\Lambda_{UV} = \sigma^{1/4}$)
- Identified that degeneracy is *relocated*, not eliminated

What was not achieved:

- N is not specified by EDC
- Regularization factor remains $\mathcal{O}(1)$ uncertainty
- σ still required (same as NC-1)

Conclusion: NC-2 does not break the degeneracy. The unknown C is replaced by unknown N . BLOCK-003 remains [OPEN].

8 Implications for BLOCK-003

Diagnosis

Both NC-1 and NC-2 fail for the same structural reason: EDC lacks a **normalization anchor**—a first-principles way to fix one dimensionless number (C , N , or equivalent).

Possible interpretations:

1. **Genuine underdetermination:** The 5D framework inherently has one free parameter that must be fixed by observation (calibration).
2. **Missing physics:** A stabilization mechanism (modulus fixing, topological quantization) is needed but not present in current EDC.
3. **Tautology:** The derivation is circular—4D gravity was presupposed somewhere in the axioms.

Recommendation: Perform tautology audit (derivation v10) before attempting further normalization candidates.